AKARSH NAVHULE

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PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 1.5+ years of industry experience building and deploying production-grade AI-powered applications. Skilled across the entire stack — React/Next.js frontends, FastAPI backends, cloud infrastructure on GCP, and AI/automation pipelines using RAG, LLMs, and n8n. Passionate about turning complex AI capabilities into practical, scalable tools.

TECHNICAL SKILLS

Languages	Python, JavaScript
Frontend	React.js, Next.js, Tailwind CSS
Backend	FastAPI, Node.js, REST API Design
AI / ML	RAG (ChromaDB), LLM APIs (Gemini, OpenAI), ComfyUI, n8n Automation, Claude Code
Databases	MongoDB, PostgreSQL, ChromaDB
Cloud & DevOps	GCP (Compute Engine, Cloud Storage), Nginx, ngrok, Shellscripts
Tools	Claude Code, Antigarvity, VS code
Prompt Eng.	Context-aware prompting with Claude Code, RAG query design, LLM instruction tuning for production outputs

WORK EXPERIENCE

Full Stack Developer · **L&T Finance** — Bengaluru, India — Jul 2024 – Oct 2025

- Built a high-throughput FastAPI service using AWS Rekognition for facial-recognition attendance, handling 500+ concurrent API calls/second.
- Architected a concurrent, multi-threaded FastAPI service with async request handling and event-loop optimizations.
- Developed a React.js analytics dashboard surfacing real-time attendance states (present / absent / face not found).
- Integrated MongoDB for persistent logging of every recognition event, enabling audit trails and reporting.
- Leveraged Claude Code with precision context-aware prompting to reduce average feature delivery from 3 weeks to 1 week.

Freelance Full Stack Developer · **Self-Employed** — Bengaluru, India — Nov 2025 – Mar 2026

- Built and maintained **Malnad Realty** (malnadrealty.com), a NoBroker-style property marketplace, integrating 150+ REST APIs across Next.js, Spring Boot, and FastAPI.
- Sole engineer for end-to-end GCP deployment: Compute Engine VMs, Nginx reverse proxy,firewall rules, MongoDB and PostgreSQL.
- Integrated OTP login, WhatsApp notification APIs, Google Cloud Storage, and headless WordPress CMS.

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PROJECTS

01 Sahayak — AI Student Assistant (Hackathon) ↗ Backend ↗ Frontend

FastAPI · Next.js · ChromaDB · Google Gemini Flash 2.5 · Google Vision API · Google Firestore · GCP VM

- Built a full-stack RAG application for textbook Q&A, question paper generation, and AI-assisted answer sheet correction.
- Designed FastAPI backend with ChromaDB vector store; integrated Google Vision API for OCR; connected Gemini Flash 2.5 for evaluation.
- Deployed both Next.js and FastAPI on a single GCP VM with Nginx reverse proxy.

02 Malnad Realty — Full-Stack Property Marketplace

Live: malnadrealty.com ↗

Next.js · Spring Boot · FastAPI · MongoDB · PostgreSQL · GCP · Nginx · WordPress CMS · Google Cloud Storage

- Integrated 150–200 REST APIs into the Next.js frontend for property listings, search, and user flows.
- Configured Nginx with multi-service routing and iptables firewall rules for security hardening.
- Integrated WhatsApp notifications, OTP login, Google Cloud Storage, and headless WordPress CMS.

03 AI Face Recognition Attendance System @ L&T Finance

FastAPI · AWS Rekognition · React.js · MongoDB · GCP VM

- Designed core FastAPI concurrency architecture enabling 500 simultaneous AWS Rekognition API calls/second.
- Built React.js analytics dashboard for real-time attendance statistics.
- Resolved multi-threading race conditions and event-loop bottlenecks via async request batching.

04 AI Automation Agents (n8n) ↗ GitHub

n8n · OpenAI GPT-4.1 · Gmail API · Razorpay · Google Calendar · Sora2 · Google Sheets

- **Auto Email Labeling & Reply Agent:** reads Gmail, classifies/labels emails, and auto-drafts replies using GPT-4.1 nano.
- **WhatsApp AI Clinic Agent:** end-to-end appointment booking via WhatsApp with Razorpay payments and Google Calendar sync.
- **AI UGC Video Generator:** product photo-to-UGC-style video pipeline using Sora2 + GPT-4.1 mini at $0.15/video.

05 Claude Code Agent Projects

Claude Code · Remotion · YouTube Data API · Google Sheets API · LLM APIs

- **YouTube Niche Analyzer Agent:** fetches top-performing videos and auto-generates Excel + PPT reports delivered via email. ↗ GitHub
- **Motion Graphics Generator:** prompt-driven motion graphics pipeline using Claude Code + Remotion, rendering to MP4. ↗ GitHub
- **Google Sheet AI Chat:** LLM-powered natural language Q&A over Google Sheets data. ↗ GitHub

06 AI Digital Twin — Face, Voice & Video Synthesis ↗ GitHub

ComfyUI · FLUX LoRA · Chatterbox (voice cloning) · Infinite Talk · Hugging Face · ngrok · Jarvis Labs GPU

- Trained a FLUX LoRA model on personal photos for consistent, high-fidelity face generation.
- Cloned voice using Chatterbox from personal audio samples and integrated it with an Infinite Talk ComfyUI workflow.
- Generated lip-synced talking video from a single image + audio + text prompt, end-to-end.

EDUCATION

B.E. — Computer Science & Engineering Jul 2020 – Jun 2024

Nitte Meenakshi Institute of Technology, Bengaluru CGPA: 8.16 / 10.0

INTERESTS & ACTIVITIES

Trekking · Long-distance bike rides · Car enthusiast · Surfing